

Multiple Choice

1. **Answer:** A

Options: A) cause particles to become neutral; B) increase the mass of the nucleus; C) cause particles to spin; D) generate heat within the nucleus; E) create magnetic fields

Note: Correct option: A - cause particles to become neutral

2. **Answer:** D

Options: A) 1000 coul; B) 1800 coul; C) 3000 coul; D) 1500 coul; E) 2500 coul

Note: Correct option: D - 1500 coul

3. **Answer:** A

Options: A) mass; B) low-frequency sound; C) nuclear power; D) energy; E) heat

Note: Correct option: A - mass

4. **Answer:** E

Options: A) 2 kT; B) (7/2) k T; C) 5 kT; D) (1/4) k T; E) 3 kT

Note: Correct option: E - 3 kT

5. **Answer:** B

Options: A) -45 ft/sec^2 and 4 sec; B) -48.4 ft/sec^2 and 4.55 sec; C) -40 ft/sec^2 and 6 sec; D) -42 ft/sec^2 and 5.5 sec; E) -52 ft/sec^2 and 4.8 sec

Note: Correct option: B - -48.4 ft/sec^2 and 4.55 sec

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '79.0 eV'.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'nucleons; electrons.'

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '500 lbs'.

Long Form Response

11. **Answer:** 4 s. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 6.5. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key